



SQL-Based Inventory Risk Analysis for Demand Planning

Using SQL to analyze demand, forecast gaps, available inventory, and WOS to identify SKU-level replenishment risk.

Tools: SQL, Excel/CSV, DB Browser for SQLite

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Project Overview

This self-directed SQL project analyzes a simulated retail product portfolio to identify potential inventory and replenishment risks.

The goal was to connect sales, forecast, product, and inventory data to answer key planning questions:

- Which SKUs are driving the highest demand?
- Which products are selling above forecast?
- Which items have low available inventory?
- Which SKUs may be at risk based on WOS?

Business Objective:

Use SQL to support demand planning decisions by identifying high-demand, low-coverage SKUs that may require replenishment review.



Dataset Structure

This project uses five connected tables to simulate a retail planning environment:

- **products:** SKU, product name, category, vendor location
- **sales_orders:** units sold, forecast quantity, channel, region
- **inventory_snapshot:** on-hand units, committed units, QA hold, safety stock
- **inbound_shipments:** inbound units, ETA, shipment status
- **vendors:** lead time, location, vendor type

Join Key: SKU was used to connect sales, forecast, product, and inventory data for SKU-level risk analysis.

SQL Analysis Process



Demand Summary

Ranked SKUs and categories by total units sold.



Forecast Gap Analysis

Compared actual units sold against forecast quantity.



Net Available Inventory

Calculated usable inventory after committed units and QA hold.



WOS Analysis

Estimated weeks of supply using net available inventory and average weekly demand.



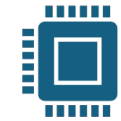
Risk Review

Identified high-demand, low-coverage SKUs for replenishment review.



SQL functions used:

JOIN, SUM, GROUP BY, ORDER BY, calculated fields



Sample SQL logic:

Joined products, sales, and inventory tables by SKU to calculate total demand, forecast gap, net available inventory, and WOS.

METRIC	KEY RESULT	PLANNING MEANING
Highest demand category	Skincare: 19,548 units	Strong category demand
Top demand SKU	Glow Serum: 8,377 units	High demand driver
Largest forecast gap	Velvet Tint: +467 units	Sold above forecast
Lowest net available	Silk Scarf: 20 units	Low availability risk
Lowest WOS	Velvet Tint: 0.05 weeks	Highest coverage risk

Key Findings

- Skincare showed the strongest category-level demand.
- Velvet Tint and Glow Serum sold above forecast.
- Low WOS indicates potential replenishment risk.

Sample SQL Query: Forecast Gap Analysis

```
SELECT
products.product_name,
products.category,
sales_orders.sku,
SUM(sales_orders.unit_sold) AS total_units_sold,
SUM(sales_orders.forecast_qty) AS total_forecast_qty,
SUM(sales_orders.unit_sold) - SUM(sales_orders.forecast_qty) AS forecast_gap
FROM sales_orders
JOIN products
ON sales_orders.sku = products.sku
GROUP BY sales_orders.sku, products.product_name, products.category
ORDER BY forecast_gap DESC;
```

This query joins product and sales data by SKU, compares actual units sold against forecast quantity, and ranks products by forecast gap.

Product	WOS
Velvet Tint	0.05
Glow Serum	0.06
Silk Scarf	0.07
Night Cream	0.40
Daily Tote	0.46
Sun Base	0.61
Slim Jeans	0.94

WOS Risk View

- **Risk Interpretation**

- All listed SKUs show less than 1 week of supply.
- Velvet Tint, Glow Serum, and Silk Scarf have the lowest coverage.
- These SKUs should be prioritized for replenishment or inbound timing review.

Lower WOS indicates lower inventory coverage and higher potential replenishment risk.

Recommendations

- **Prioritize replenishment review for low-WOS SKUs**
Velvet Tint, Glow Serum, and Silk Scarf should be reviewed first due to low inventory coverage.
- **Monitor SKUs selling above forecast**
Products with positive forecast gaps should be monitored closely to avoid potential stockout risk.
- **Review inbound timing and supply availability**
Low-coverage SKUs should be checked against inbound shipments, vendor lead times, and shipment status.
- **Use inventory risk flags for planning decisions**
Combining forecast gap, net available inventory, and WOS can help planners identify replenishment priorities earlier.
- **Next step**
Add automated risk flags and dashboard visuals to make the analysis easier to review by SKU, category, and vendor.

Key Takeaway:

SQL can support planning decisions by connecting demand, forecast, and inventory data into one risk-based view.